Assignment 2 Report

Building a Structure-from-Motion Pipeline

Adit Jindal - 2023CS50626

Contents

1 Introduction **2**

2 Standard Convolutional Neural Networks - Task 1.1 **2**
2.1 Introduction 2
2.2 Training 2
2.3 Testing 4
2.4 Ablation 4

3 Enhanced Convolutional Neural Networks - Task 1.2 **5**
3.1 Introduction 5
3.2 Training 6
3.3 Testing 7
3.4 Ablation 8

4 Standard Vision Transformer - Task 2.1 **9**
4.1 Introduction 9
4.2 Training 9
4.3 Testing 11
4.4 Ablation 11

5 Vision Transformer without Normalization - Task 2.2 **13**
5.1 Introduction 13
5.2 Training 13
5.3 Testing 15
5.4 Ablation 15

6 Grad-CAM Visualizations - 3.1 **16**
6.1 Grad-CAM Analysis 16

7 Attention Maps - 3.2 **20**
7.1 Attention Map Analysis (DeiT-3 Small) 20

1 Introduction

This assignment involves multiclass land-use classification using Sentinel-2 satellite RGB images, using two deep learning approaches: Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), tasks 1 and 2 respectively.

2 Standard Convolutional Neural Networks - Task 1.1

2.1 Introduction

In this task, we build a multiclass image classification pipeline using a pretrained ResNet-18 model and fine-tune it for a 10-class land-use classification problem. The model is adapted by modifying its final layer to match the number of classes and trained using RGB satellite images.

We train the model using standard settings and evaluate its performance using metrics like accuracy, macro F1-score, and ROC-AUC. The best model is selected based on validation performance.

Additionally, an ablation study is conducted by replacing the standard loss function with Focal Loss to observe its impact on performance.

2.2 Training

File used : `train11.py`

In this section, we analyze the performance of the ResNet-18 model across training epochs for the multiclass land-use classification task using RGB satellite images.

Table below shows the performance metrics across 10 epochs.

Epoch	Train Loss	Accuracy	Macro F1-score	ROC-AUC
1	0.2175	0.9426	0.9405	0.9990
2	0.0833	0.9569	0.9562	0.9991
3	0.0526	0.9711	0.9701	0.9994
4	0.0387	0.9756	0.9746	0.9995
5	0.0286	0.9774	0.9765	0.9995
6	0.0339	0.9678	0.9667	0.9995
7	0.0239	0.9728	0.9721	0.9995
8	0.0247	0.9739	0.9729	0.9995
9	0.0190	0.9739	0.9733	0.9994
10	0.0206	0.9793	0.9785	0.9994

Table 1: Performance metrics across epochs

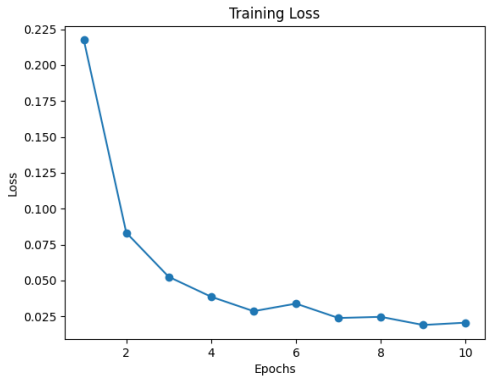

Figure 1: Training Loss vs Epochs

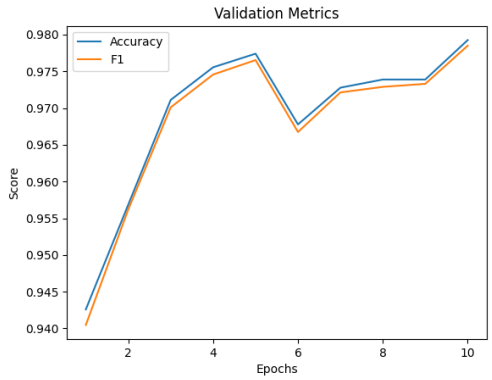

Figure 2: Macro F1-score vs Epochs

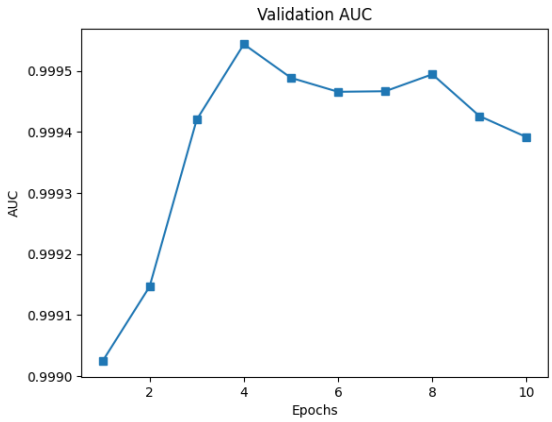

Figure 3: ROC-AUC vs Epochs

Trends observed:

- The training loss shows a clear decreasing trend, dropping significantly from 0.2175 in the first epoch to around 0.02 by the final epochs.
- Accuracy and Macro F1-score both improve steadily during the initial epochs, with rapid gains up to epoch 4–5. After this point, the performance begins to plateau, with only marginal improvements and slight fluctuations. A small dip is observed at epoch 6, which may indicate minor instability or overfitting, but the model quickly recovers in subsequent epochs.
- The ROC-AUC score remains consistently very high (above 0.999) across all epochs, suggesting that the model is highly effective at distinguishing between classes. However, since AUC saturates early, it becomes less sensitive to improvements compared to Accuracy and F1-score.
- The best overall performance is achieved at epoch **10**, where the model reaches the highest Accuracy (0.9793) and Macro F1-score (0.9785). Although the ROC-AUC is slightly higher at earlier epochs, the difference is negligible, and the final epoch provides the best balance across all evaluation metrics.

Overall, the results demonstrate that the fine-tuned ResNet-18 model performs very well on the multiclass land-use classification task, achieving high accuracy and strong class-balanced performance. The early saturation of AUC and the plateauing of other metrics suggest that the model converges quickly, and further improvements may require architectural changes or alternative loss functions.

2.3 Testing

File used: `test11.py`

Running the model trained above on the test data, we get the following results:

Accuracy : 0.9885

F1 score : 0.988

Macro ROC AUC : 0.9997

Configurations for the model trained:

Model: ResNet-18

Learning Rate: 0.0001

Epochs: 10

Batch size": 32

2.4 Ablation

File used: `ablation11.py`

In this experiment, we replace the standard CrossEntropyLoss with Focal Loss to study its effect on model performance.

Focal Loss is designed to address class imbalance by reducing the contribution of easy-to-classify examples and focusing more on hard, misclassified samples. While CrossEntropyLoss treats all samples equally, Focal Loss introduces a modulating factor that down-weights well-classified examples, allowing the model to focus more on informative and difficult samples.

Mathematically, Focal Loss adds a factor $(1 - p_t)^\gamma$ to the standard cross-entropy loss, where p_t is the predicted probability for the true class.

The parameter γ (gamma) controls the strength of this modulation:

- $\gamma = 0$ reduces Focal Loss to standard CrossEntropyLoss
- Higher values of γ increase focus on harder-to-classify examples

In this experiment, we evaluate $\gamma \in \{1, 2, 3\}$ to observe its effect on performance. These values are chosen to provide a reasonable range of focusing strength while maintaining training stability. (For efficiency, only 3 training runs were performed per γ, as very minimal variation was observed across additional runs.)

γ	Best ROC-AUC
1	0.999355
2	0.999350
3	0.999450

Table 2: Best ROC-AUC for different values of γ

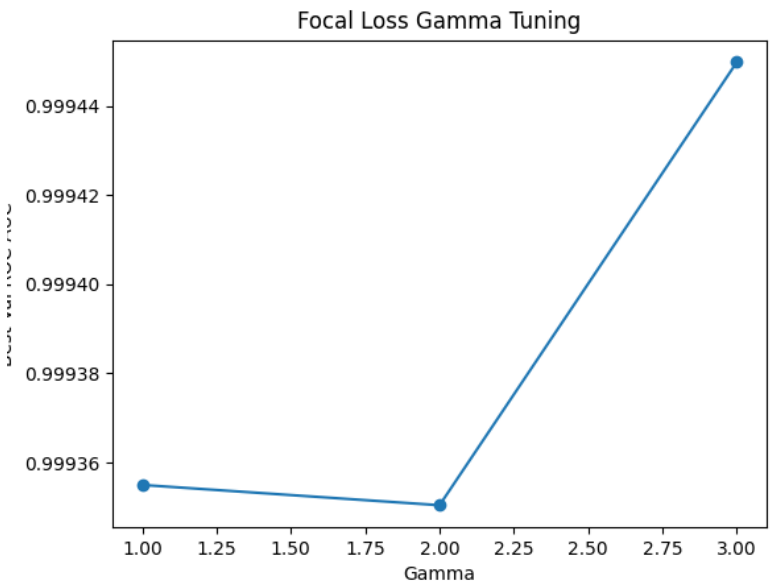

Figure 4: Best ROC-AUC vs γ

From Table 8 and Figure 16, we observe that the performance remains extremely stable across different values of γ. All configurations achieve very high ROC-AUC scores (around 0.9993–0.9994), indicating that the model already performs strongly even without strong focal adjustment.

Among the tested values, $\gamma = 3$ achieves the highest ROC-AUC (0.99945), but the improvement over $\gamma = 1$ and $\gamma = 2$ is extremely marginal (graph is scaled up to magnify the difference). This suggests that increasing the focusing strength does not significantly impact model discrimination ability for this dataset.

This behavior indicates that the dataset is likely not heavily affected by class imbalance, or that the baseline ResNet-18 model already learns robust feature representations where hard examples are not dominant. As a result, **Focal Loss does not provide a substantial advantage** over standard CrossEntropyLoss in this setting.

3 Enhanced Convolutional Neural Networks - Task 1.2

3.1 Introduction

We enhance the previously used ResNet-18 model by integrating Squeeze-and-Excitation (SE) blocks to improve feature representation. SE blocks help the network focus on more informative feature channels by learning channel-wise importance weights through

a squeeze-and-excitation mechanism.

We insert SE blocks after each residual stage of ResNet-18 (layer1 to layer4) and fine-tune the modified network for the same 10-class land-use classification task. The training setup remains unchanged from the previous experiment.

Finally, we compare the results with the baseline ResNet-18 model and also perform an ablation study using Focal Loss to analyze its effect in combination with SE-enhanced features.

3.2 Training

File used : `train12.py`

In this section, we analyze the performance of the SE-ResNet-18 model across training epochs for the multiclass land-use classification task using RGB satellite images.

Table below shows the performance metrics across 10 epochs.

Epoch	Train Loss	Accuracy	Macro F1-score	ROC-AUC
1	0.2541	0.9713	0.9703	0.9994
2	0.0856	0.9737	0.9728	0.9996
3	0.0537	0.9759	0.9745	0.9994
4	0.0408	0.9757	0.9751	0.9996
5	0.0360	0.9696	0.9688	0.9994
6	0.0304	0.9807	0.9800	0.9997
7	0.0199	0.9744	0.9735	0.9996
8	0.0230	0.9750	0.9742	0.9995
9	0.0251	0.9754	0.9747	0.9996
10	0.0194	0.9748	0.9737	0.9996

Table 3: Performance metrics across epochs

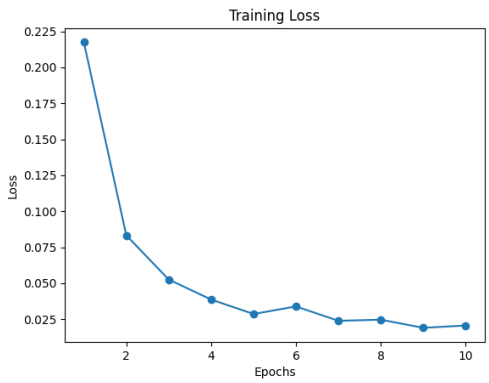

Figure 5: Training Loss vs Epochs

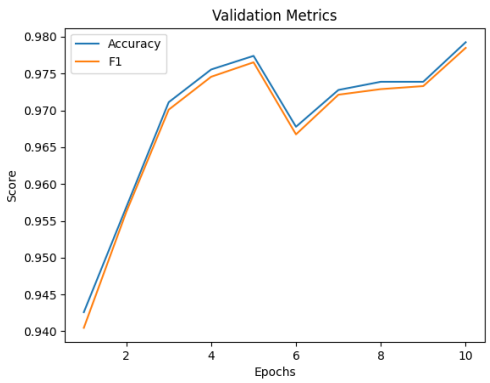

Figure 6: Macro F1-score vs Epochs

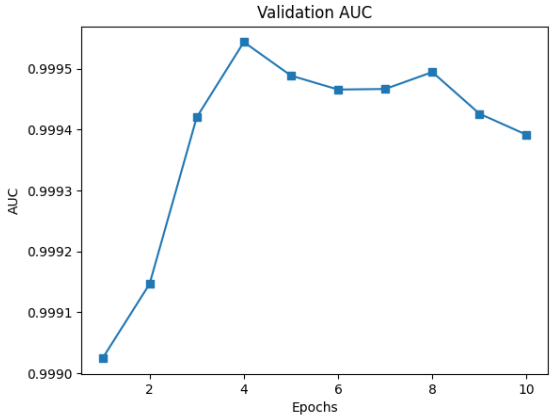

Figure 7: ROC-AUC vs Epochs

Trends observed:

- The training loss shows a clear decreasing trend, dropping significantly from 0.2541 in the first epoch to around 0.02 by the final epochs. This indicates stable convergence of the SE-enhanced model.
- Accuracy and Macro F1-score remain consistently high across all epochs, with minor fluctuations. A slight performance drop is observed around epoch 5, followed by recovery in later epochs. Overall, the model maintains strong and stable performance throughout training.
- The ROC-AUC score remains extremely high (around 0.9994–0.9997) across all epochs, with very small variation. This indicates that the model is highly effective at separating classes and that SE blocks do not destabilize discrimination performance.
- The best overall performance is achieved at epoch **6**, where the model reaches the highest Accuracy (0.9807) and Macro F1-score (0.9800). This suggests that the SE-ResNet converges slightly earlier compared to the baseline model.

Overall, the results demonstrate that the incorporation of Squeeze-and-Excitation blocks improves the stability and representational power of the ResNet-18 model. Compared to the baseline, the SE-enhanced model achieves slightly better peak performance and more consistent learning behavior across epochs. The early saturation of ROC-AUC further suggests that the model is highly confident in class separation for this dataset.

3.3 Testing

File used: `test12.py`

Running the model trained above on the test data, we get the following results:

Accuracy : 0.9822

F1 score : 0.9821

Macro ROC AUC : 0.9992
Configurations for the model trained:

Model: SE-ResNet-18

Learning Rate: 0.0001

Epochs: 10

Batch size: 32

3.4 Ablation

File used: `ablation12.py`

In this experiment, we replace the standard CrossEntropyLoss with Focal Loss to study its effect on model performance.

Focal Loss is designed to address class imbalance by reducing the contribution of easy-to-classify examples and focusing more on hard, misclassified samples. While CrossEntropyLoss treats all samples equally, Focal Loss introduces a modulating factor that down-weights well-classified examples, allowing the model to focus more on informative and difficult samples.

Mathematically, Focal Loss adds a factor $(1 - p_t)^\gamma$ to the standard cross-entropy loss, where p_t is the predicted probability for the true class.

The parameter γ (gamma) controls the strength of this modulation:

- $\gamma = 0$ reduces Focal Loss to standard CrossEntropyLoss
- Higher values of γ increase focus on harder-to-classify examples

In this experiment, we evaluate $\gamma \in \{1, 2, 3\}$ to observe its effect on performance. These values are chosen to provide a reasonable range of focusing strength while maintaining training stability. (For efficiency, only 3 training runs were performed per γ, as very minimal variation was observed across additional runs.)

γ	Best ROC-AUC
1	0.999355
2	0.999350
3	0.999450

Table 4: Best ROC-AUC for different values of γ

From Table 8 and Figure 16, we observe that the performance remains extremely stable across different values of γ. All configurations achieve very high ROC-AUC scores (around 0.9993–0.9994), indicating that the model already performs strongly even without strong focal adjustment.

Among the tested values, $\gamma = 3$ achieves the highest ROC-AUC (0.99945), but the improvement over $\gamma = 1$ and $\gamma = 2$ is extremely marginal (graph is scaled up to magnify the difference). This suggests that increasing the focusing strength does not significantly impact model discrimination ability for this dataset.

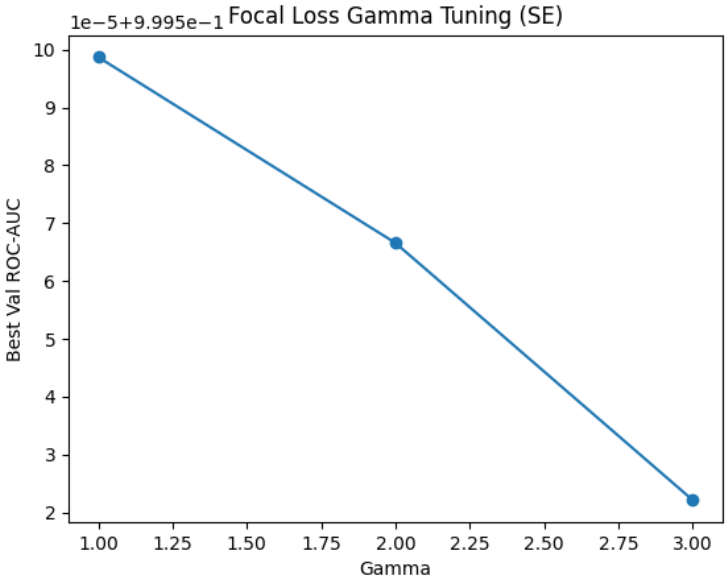

Figure 8: Best ROC-AUC vs γ

This behavior indicates that the dataset is likely not heavily affected by class imbalance, or that the baseline ResNet-18 model already learns robust feature representations where hard examples are not dominant. As a result, **Focal Loss does not provide a substantial advantage** over standard CrossEntropyLoss in this setting.

4 Standard Vision Transformer - Task 2.1

4.1 Introduction

In this subtask, we explore transformer-based architectures for image classification by fine-tuning a pretrained DeiT-3 Small model. The model is initialized with ImageNet pretrained weights and adapted for the 10-class land-use classification task using RGB satellite images.

We use the CLS token representation for final classification and replace the original head with a task-specific layer. The model is trained using the same setup as earlier tasks and evaluated using Accuracy, Macro F1-score, and ROC-AUC. Finally, we compare its performance with the CNN-based models from Task 1 and also perform an ablation study using Focal Loss.

4.2 Training

File used : `train21.py`
In this section, we analyze the performance of the Vision Transformer (DeiT-3 Small) model across training epochs for the multiclass land-use classification task using RGB satellite images.

Table below shows the performance metrics across 10 epochs.

Epoch	Train Loss	Accuracy	Macro F1-score	ROC-AUC
1	0.1813	0.9680	0.9667	0.9993
2	0.0602	0.9746	0.9736	0.9996
3	0.0420	0.9722	0.9716	0.9996
4	0.0449	0.9739	0.9724	0.9995
5	0.0301	0.9781	0.9772	0.9995
6	0.0243	0.9780	0.9774	0.9995
7	0.0221	0.9572	0.9572	0.9991
8	0.0192	0.9761	0.9754	0.9994
9	0.0288	0.9689	0.9681	0.9995
10	0.0150	0.9815	0.9807	0.9995

Table 5: Performance metrics across epochs

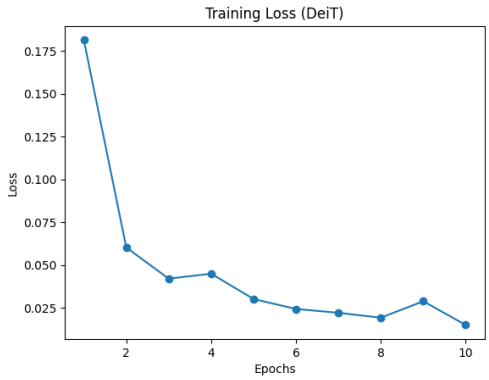

Figure 9: Training Loss vs Epochs

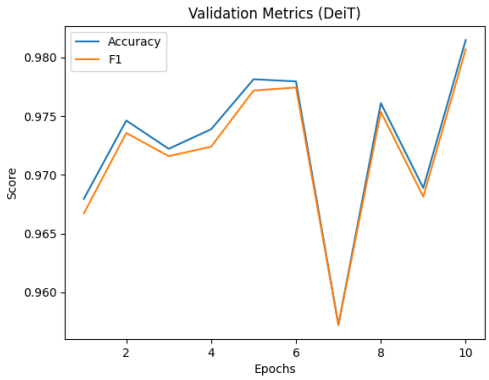

Figure 10: Macro F1-score vs Epochs

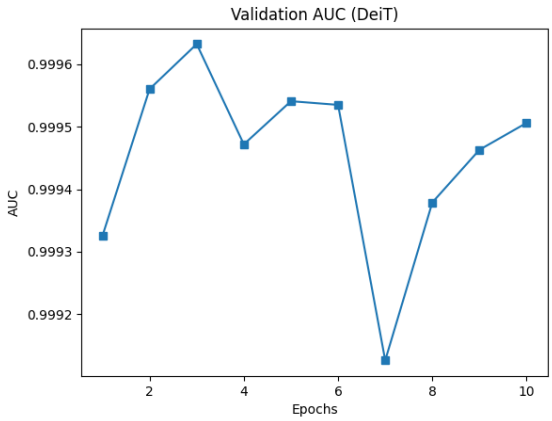

Figure 11: ROC-AUC vs Epochs

Trends observed:

- The training loss shows a consistent decreasing trend, dropping from 0.1813 in the first epoch to 0.0150 by the final epoch, indicating stable convergence of the transformer model.

- Accuracy and Macro F1-score improve overall across epochs, though with some fluctuations. A noticeable drop is observed at epoch 7, followed by recovery in later epochs, suggesting temporary instability during training.
- The ROC-AUC score remains extremely high (around 0.9991–0.9996) across all epochs, indicating strong class separation capability. However, similar to earlier models, it saturates early and shows limited sensitivity to incremental improvements.
- The best overall performance is achieved at epoch **10**, where the model attains the highest Accuracy (0.9815) and Macro F1-score (0.9807), making it the most effective checkpoint.

Overall, the Vision Transformer (DeiT-3 Small) demonstrates strong performance on the land-use classification task, achieving slightly better peak accuracy and F1-score compared to the CNN-based models. While performance remains stable overall, minor fluctuations suggest that transformer-based models may require more careful tuning to ensure consistent training stability across epochs.

4.3 Testing

File used: `test21.py`
Running the model trained above on the test data, we get the following results:

Accuracy : 0.9804

F1 score : 0.9797

Macro ROC AUC : 0.9995
Configurations for the model trained:

Model: DeiT-3 Small

Learning Rate: 0.0001

Epochs: 10

Batch size: 32

4.4 Ablation

File used: `ablation21.py`
In this experiment, we replace the standard CrossEntropyLoss with Focal Loss to study its effect on model performance.

Focal Loss is designed to address class imbalance by reducing the contribution of easy-to-classify examples and focusing more on hard, misclassified samples. While CrossEntropyLoss treats all samples equally, Focal Loss introduces a modulating factor that down-weights well-classified examples, allowing the model to focus more on informative and difficult samples.

Mathematically, Focal Loss adds a factor $(1 - p_t)^\gamma$ to the standard cross-entropy loss, where p_t is the predicted probability for the true class.

The parameter γ (gamma) controls the strength of this modulation:

- $\gamma = 0$ reduces Focal Loss to standard CrossEntropyLoss
- Higher values of γ increase focus on harder-to-classify examples

In this experiment, we evaluate $\gamma \in \{1, 2, 3\}$ to observe its effect on performance. These values are chosen to provide a reasonable range of focusing strength while maintaining training stability. (For efficiency, only 3 training runs were performed per γ, as very minimal variation was observed across additional runs.)

γ	Best ROC-AUC
1	0.999652
2	0.999587
3	0.999618

Table 6: Best ROC-AUC for different values of γ

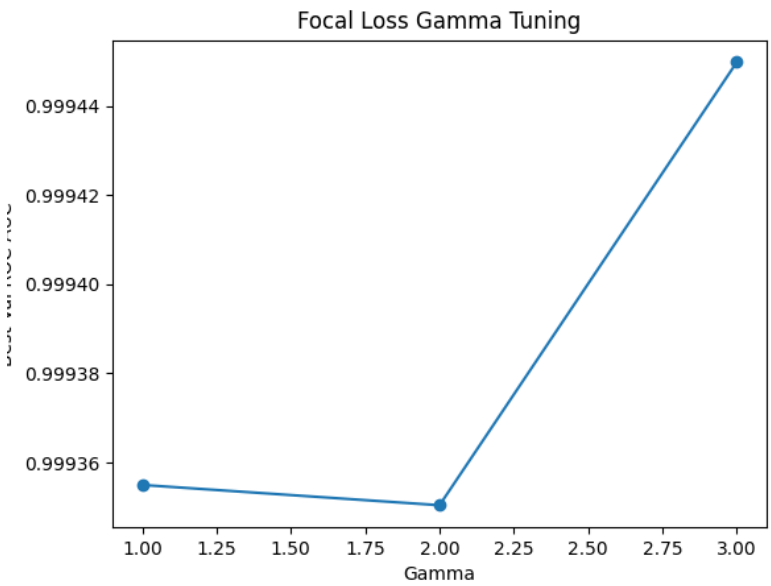

Figure 12: Best ROC-AUC vs γ

From Table 8 and Figure 16, we observe that the performance remains extremely stable across different values of γ. All configurations achieve very high ROC-AUC scores (around 0.9996), indicating that the model already performs strongly even without strong focal adjustment.

Among the tested values, $\gamma = 1$ achieves the highest ROC-AUC (0.999652), but the differences across $\gamma = 2$ and $\gamma = 3$ are extremely marginal. This suggests that increasing the focusing strength does not significantly impact the model's discrimination ability for this dataset.

This behavior indicates that the dataset is likely not heavily affected by class imbalance, or that the DeiT-3 Small model already learns highly robust representations

where hard examples are not dominant. As a result, **Focal Loss does not provide a substantial advantage** over standard CrossEntropyLoss in this setting.

5 Vision Transformer without Normalization - Task 2.2

5.1 Introduction

In this subtask, we explore a recent modification to Transformer architectures by replacing Layer Normalization with Dynamic Tanh (DyT). DyT uses a simple learnable tanh-based transformation to stabilize activations without explicit normalization.

We modify the previously used DeiT-3 Small model by replacing its LayerNorm layers with DyT and fine-tune it for the same 10-class land-use classification task. The model is trained under the same setup as before and evaluated using Accuracy, Macro F1-score, and ROC-AUC.

Finally, we compare its performance with the standard Vision Transformer and also conduct an ablation study using Focal Loss.

5.2 Training

File used : `train22.py`

In this section, we analyze the performance of the DyT-Transformer (DeiT-3 Small with Dynamic Tanh) model across training epochs for the multiclass land-use classification task using RGB satellite images.

Table below shows the performance metrics across 10 epochs.

Epoch	Train Loss	Accuracy	Macro F1-score	ROC-AUC
1	1.0948	0.7737	0.7580	0.9691
2	0.5696	0.8422	0.8361	0.9851
3	0.4106	0.8893	0.8862	0.9932
4	0.2967	0.8963	0.8910	0.9925
5	0.2537	0.9209	0.9165	0.9959
6	0.2052	0.9237	0.9217	0.9963
7	0.1870	0.9331	0.9302	0.9967
8	0.1603	0.9409	0.9384	0.9977
9	0.1534	0.9359	0.9338	0.9964
10	0.1282	0.9383	0.9359	0.9968

Table 7: Performance metrics across epochs

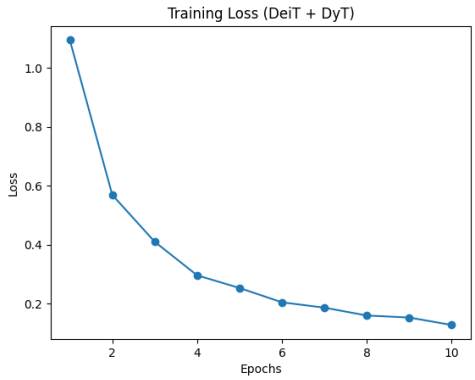

Figure 13: Training Loss vs Epochs

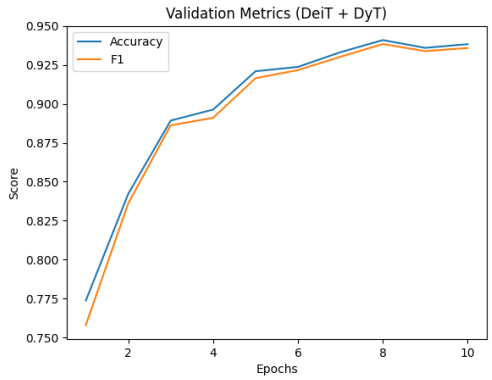

Figure 14: Macro F1-score vs Accuracy

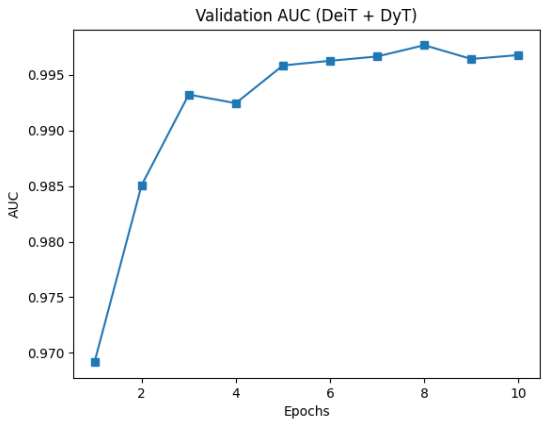

Figure 15: ROC-AUC vs Epochs

Trends observed:

- The training loss shows a strong and consistent decreasing trend, dropping from 1.0948 in the first epoch to 0.1282 by the final epoch. This indicates stable convergence, although slower compared to standard Vision Transformers.
- Accuracy and Macro F1-score improve significantly in the early epochs, with rapid gains up to around epoch 3–5. After this point, the model continues to improve gradually but begins to plateau, with minor fluctuations in later epochs.
- The ROC-AUC score steadily increases from 0.9691 to around 0.996–0.997, showing that the model learns strong class separation capabilities. However, it remains slightly lower and more variable compared to the standard ViT model, indicating reduced calibration stability.
- The best overall performance is achieved around epoch **8**, where the model reaches the highest Accuracy (0.9409) and competitive Macro F1-score (0.9384). Later epochs show slight degradation or saturation, suggesting early convergence.

Overall, the DyT-based Transformer demonstrates moderate performance on the land-use classification task. While it is able to learn meaningful representations, its final accuracy and F1-score are lower than the standard Vision Transformer, suggesting that replacing LayerNorm with Dynamic Tanh may reduce optimization stability in this setting, despite competitive ROC-AUC performance.

5.3 Testing

File used: `test22.py`
Running the model trained above on the test data, we get the following results:

Accuracy : 0.9433

F1 score : 0.9420

Macro ROC AUC : 0.9976
Configurations for the model trained:

Model: DeiT-3 Small + DyT (No LayerNorm)

Learning Rate: 0.0001

Epochs: 10

Batch size: 32

5.4 Ablation

File used: `ablation22.py`

In this experiment, we replace the standard CrossEntropyLoss with Focal Loss to study its effect on model performance.

Focal Loss is designed to address class imbalance by reducing the contribution of easy-to-classify examples and focusing more on hard, misclassified samples. While CrossEntropyLoss treats all samples equally, Focal Loss introduces a modulating factor that down-weights well-classified examples, allowing the model to focus more on informative and difficult samples.

Mathematically, Focal Loss adds a factor $(1 - p_t)^\gamma$ to the standard cross-entropy loss, where p_t is the predicted probability for the true class.

The parameter γ (gamma) controls the strength of this modulation:

- $\gamma = 0$ reduces Focal Loss to standard CrossEntropyLoss
- Higher values of γ increase focus on harder-to-classify examples

In this experiment, we evaluate $\gamma \in \{1, 2, 3\}$ to observe its effect on performance. These values are chosen to provide a reasonable range of focusing strength while maintaining training stability. (For efficiency, only 3 training runs were performed per γ, as very minimal variation was observed across additional runs.)

γ	Best ROC-AUC
1	0.9941
2	0.9891
3	0.9922

Table 8: Best ROC-AUC for different values of γ

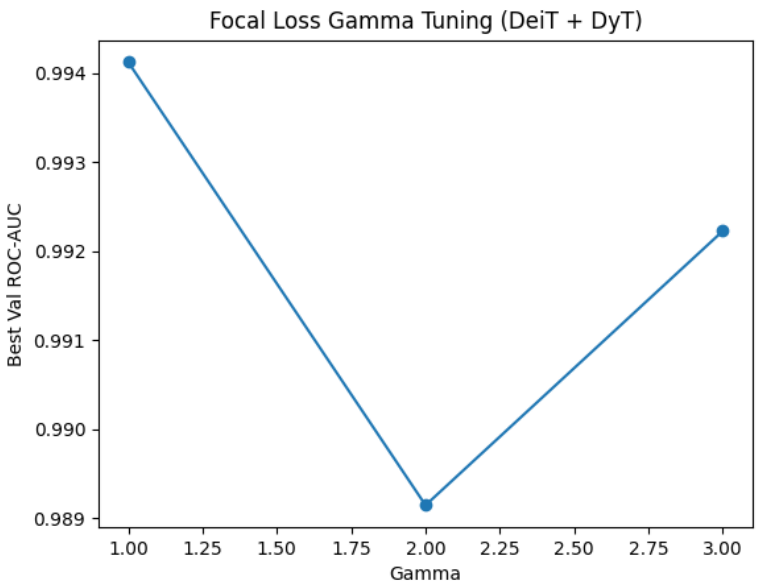

Figure 16: Best ROC-AUC vs γ

From Table 8 and Figure 16, we observe that the performance varies slightly across different values of γ, unlike earlier models where the differences were negligible. Here, $\gamma = 1$ achieves the best ROC-AUC (0.9941), while performance drops noticeably at $\gamma = 2$ before partially recovering at $\gamma = 3$.

This indicates that increasing the focusing strength does not consistently improve performance for the DyT-based Transformer. In fact, the instability introduced by higher γ values may suggest that the model is already sensitive to optimization changes due to the absence of LayerNorm, making it less robust to additional loss reweighting.

Overall, these results suggest that **Focal Loss does not provide consistent benefits** for the DyT-based Transformer and may even slightly degrade performance depending on the choice of γ. This further highlights that normalization-free Transformer variants may require more careful training dynamics and hyperparameter tuning compared to standard Vision Transformers.

6 Grad-CAM Visualizations - 3.1

6.1 Grad-CAM Analysis

To better understand the decision-making behavior of the trained models, we visualize Grad-CAM heatmaps for sample images from each class in the test set. We compare the standard ResNet-18 model with the SE-augmented ResNet-18 model to analyze how channel-wise attention affects spatial focus.

Overall, both models attend to semantically meaningful regions; however, the SE-ResNet model often produces more focused and localized activation maps.

Class-wise Observations

Class 0 The standard ResNet-18 model activates relatively broad regions including background areas, whereas the SE-ResNet model shows slightly more focused attention on the discriminative region, though the difference is not very pronounced.

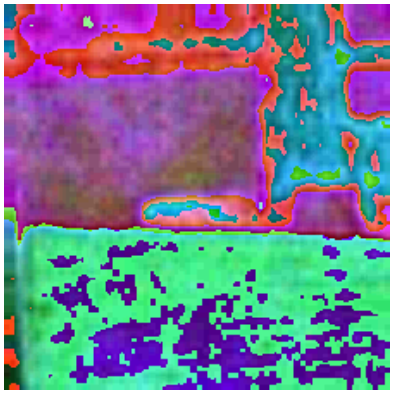

Figure 17: Original

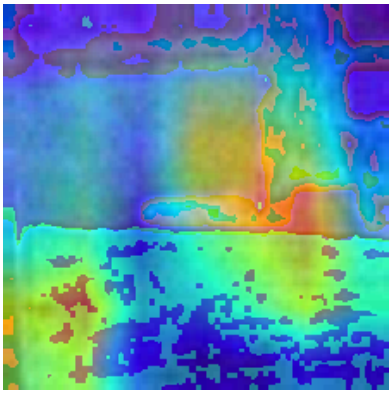

Figure 18: ResNet-18

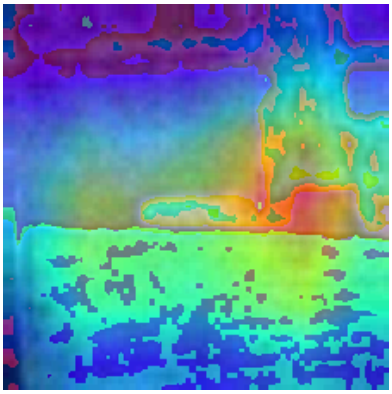

Figure 19: SE-ResNet

Class 1 ResNet-18 shows more dispersed attention, while SE blocks improve focus on the most relevant region, leading to better localization.

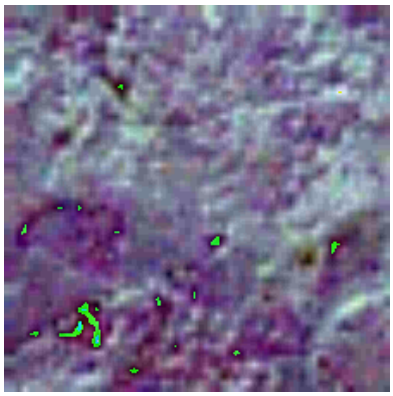

Figure 20: Original

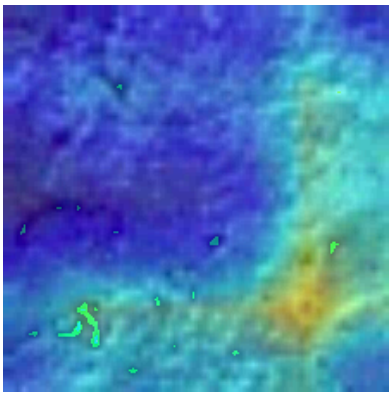

Figure 21: ResNet-18

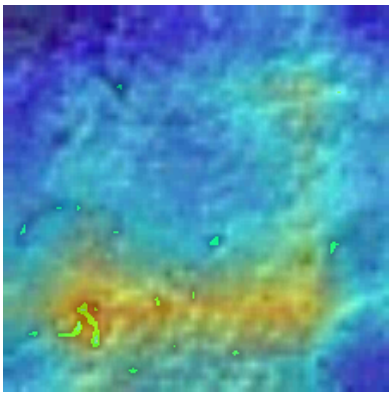

Figure 22: SE-ResNet

Class 2 Both models correctly identify the main discriminative region with minimal differences in attention distribution.

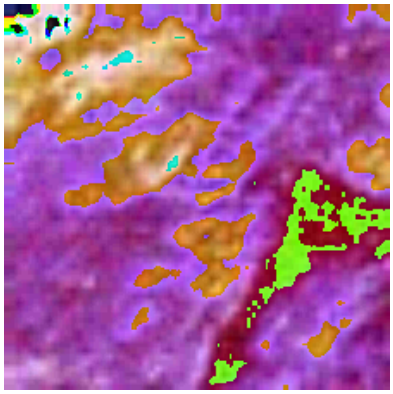

Figure 23: Original

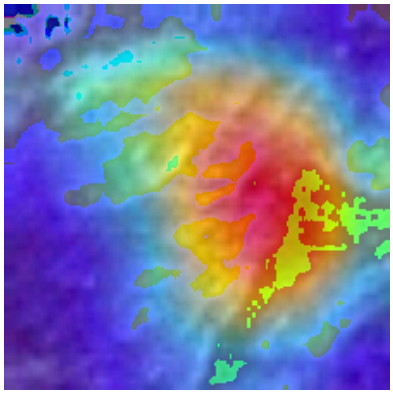

Figure 24: ResNet-18

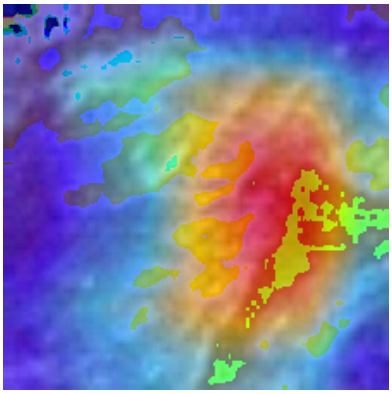

Figure 25: SE-ResNet

Class 3 The baseline model attends to only parts of the relevant region, whereas SE-ResNet shows more complete coverage of discriminative spatial features.

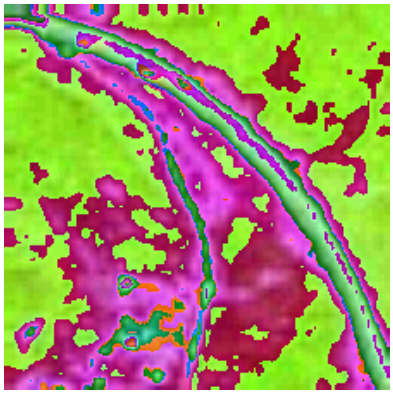

Figure 26: Original

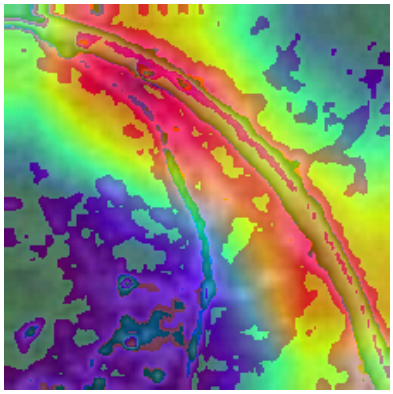

Figure 27: ResNet-18

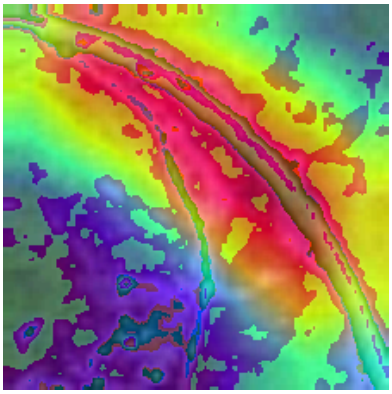

Figure 28: SE-ResNet

Class 4 Both models focus on nearly identical regions, indicating strong and stable feature learning.

Figure 29: Original

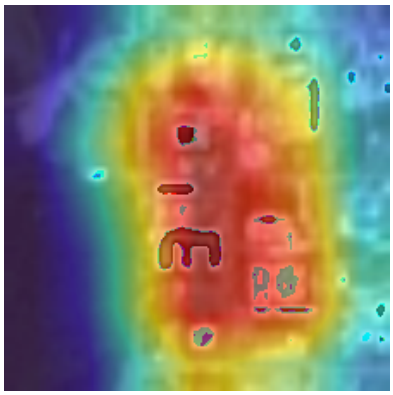

Figure 30: ResNet-18

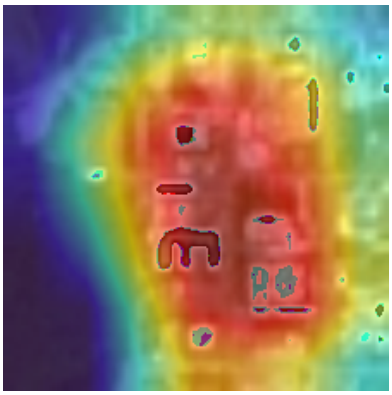

Figure 31: SE-ResNet

Class 5 The standard ResNet-18 model tends to activate a relatively broad region, including some background areas. In contrast, the SE-ResNet model focuses more precisely on the central discriminative region, leading to slightly improved localization.

Figure 32: Original

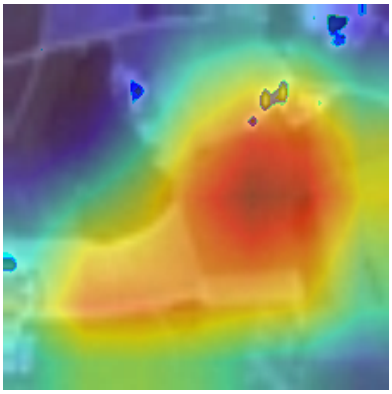

Figure 33: ResNet-18

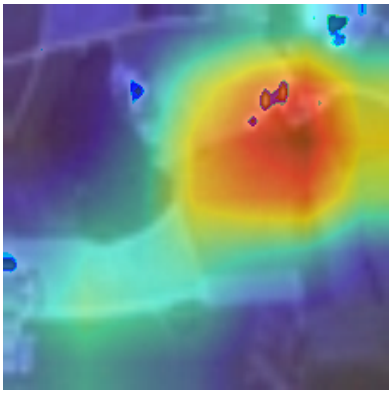

Figure 34: SE-ResNet

Class 6 For this class, both models identify multiple relevant regions across the image. However, the SE-ResNet model produces more localized and concentrated attention compared to the baseline, which appears more diffused.

Figure 35: Original

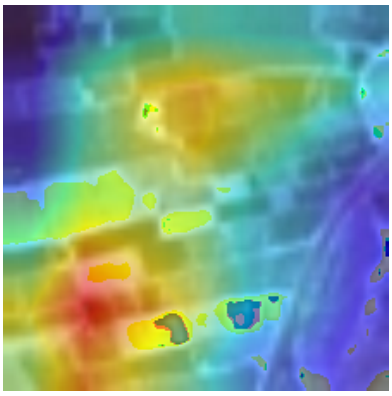

Figure 36: ResNet-18

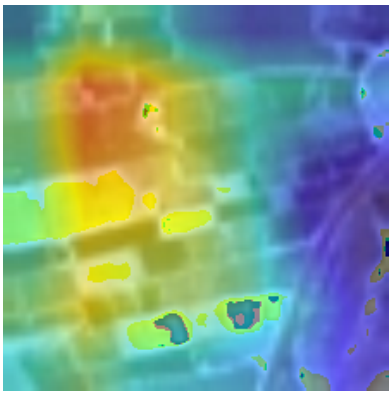

Figure 37: SE-ResNet

Class 7 Both models attend to large portions of the image, indicating that the discriminative features are widely distributed. The SE-ResNet model shows slightly more structured and less noisy activation patterns compared to the baseline.

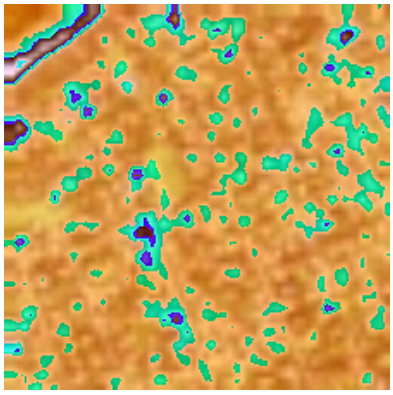

Figure 38: Original

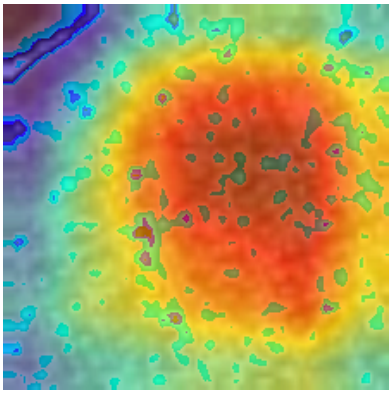

Figure 39: ResNet-18

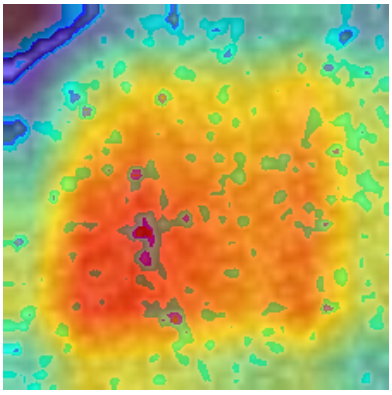

Figure 40: SE-ResNet

Class 8 In this case, the SE-ResNet model highlights a broader but more consistent region of interest compared to the baseline model, which focuses on a subset of the

relevant area. Importantly, both models still correctly capture the discriminative region.

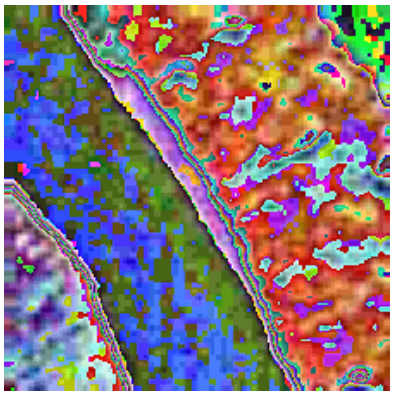

Figure 41: Original

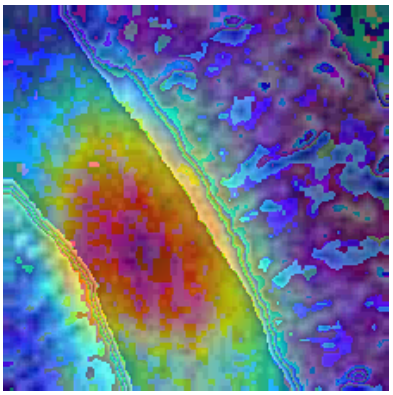

Figure 42: ResNet-18

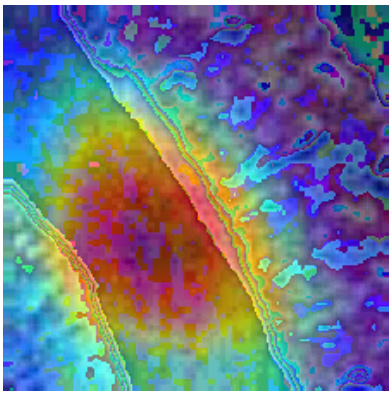

Figure 43: SE-ResNet

Class 9 The same trend is observed here, but more strongly. The SE-ResNet model activates a larger and more continuous region of interest, while the standard ResNet-18 model focuses on a smaller subset. Since the selected region is still semantically meaningful, the SE-based broader attention does not negatively affect performance.

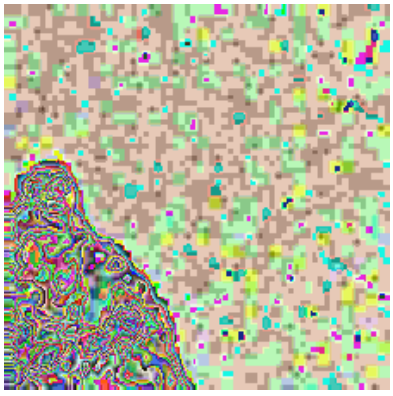

Figure 44: Original

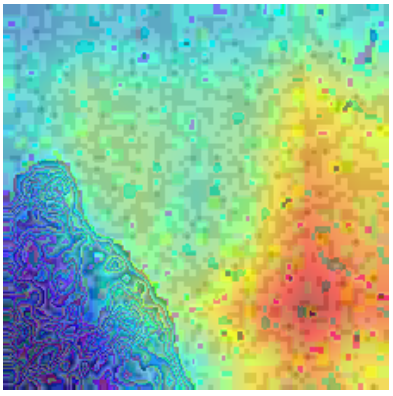

Figure 45: ResNet-18

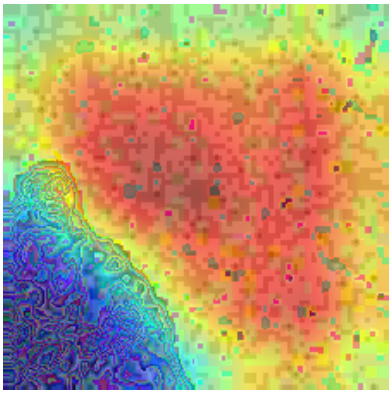

Figure 46: SE-ResNet

Overall, the SE-augmented model demonstrates improved attention consistency and slightly better localization of discriminative regions. This indicates that channel-wise recalibration via SE blocks enhances feature selectivity and improves interpretability without altering the final prediction behavior significantly.

7 Attention Maps - 3.2

7.1 Attention Map Analysis (DeiT-3 Small)

To understand the decision-making behavior of the Vision Transformer model, we visualize attention maps extracted from the final Transformer encoder block. Specifically, we use the CLS token attention over image patches to identify regions influencing the final classification decision.

Class-wise Observations

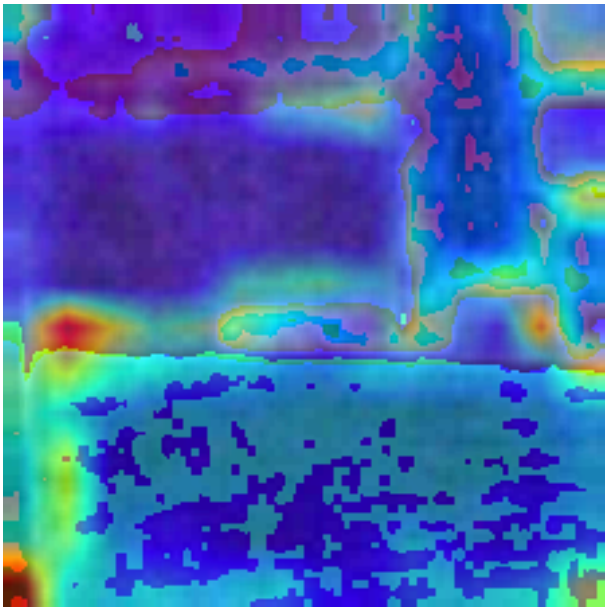

Figure 47: CLS token attention map for Class 0

Class 0 Analysis:

The attention map shows that the model focuses on the most discriminative region of the image, the edges and vertices, with moderate spread across surrounding areas.

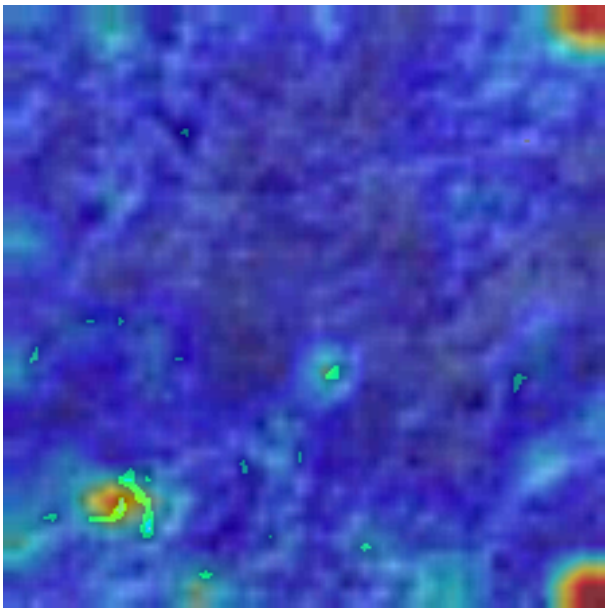

Figure 48: CLS token attention map for Class 1

Class 1 Analysis:

The model attends primarily to sparsely psread out regions, mainly clustersof trees. Background regions receive relatively low attention.

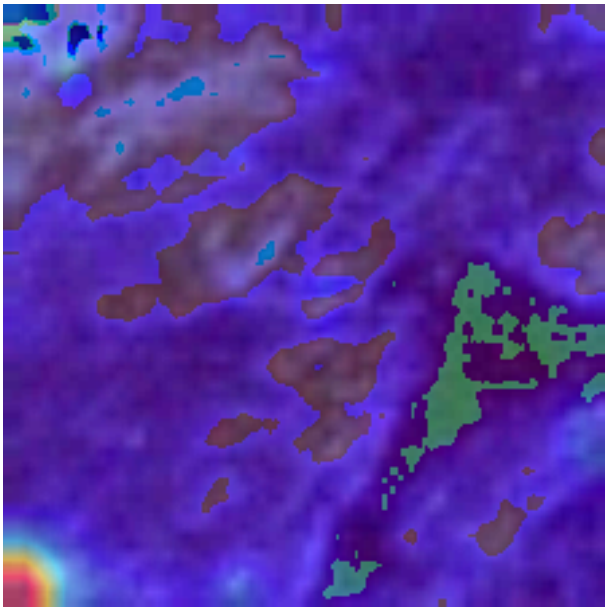

Figure 49: CLS token attention map for Class 2

Class 2 Analysis:

The attention is distributed across multiple meaningful regions with different intensities, suggesting that the model relies on a combination of spatial cues for classification.

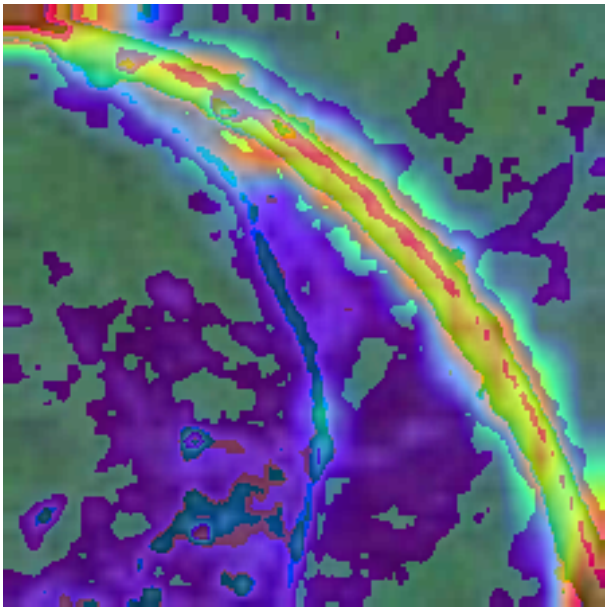

Figure 50: CLS token attention map for Class 3

Class 3 Analysis:

The model focuses strongly on a main highway, with minimal attention given to background areas. This indicates strong localization and effective feature extraction for this class.

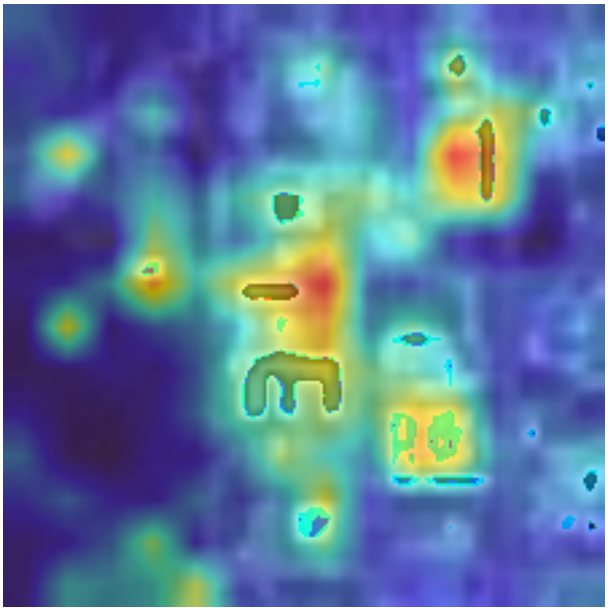

Figure 51: CLS token attention map for Class 4

Class 4 Analysis:

The attention map is correctly differentiates different rooftops at multiple regions. The model appears to rely on distributed spatial cues rather than a single dominant region.

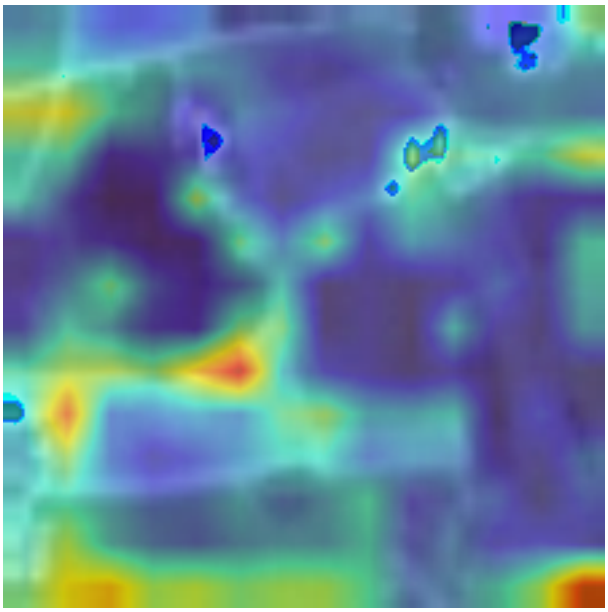

Figure 52: CLS token attention map for Class 5

Class 5 Analysis:

The model attends strongly to boundaries of pastures, since they are the only major features in the image.

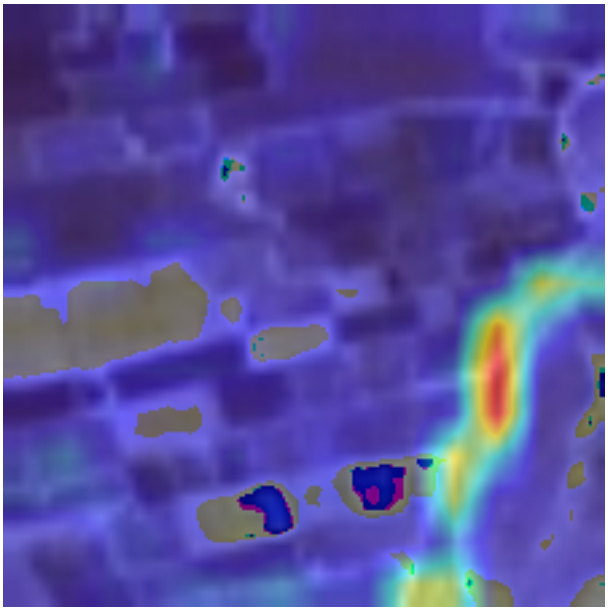

Figure 53: CLS token attention map for Class 6

Class 6 Analysis:

The attention is more distributed across the image in teh distributed sections of cropping regions, correctly recognising the rectangular areas.

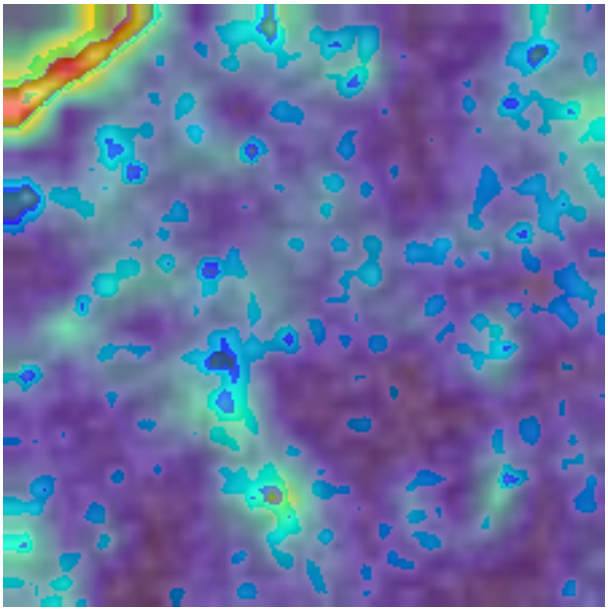

Figure 54: CLS token attention map for Class 7

Class 7 Analysis:

The model exhibits broad attention across the image, suggesting that discriminative features are widely distributed. It is not able to point to any specific area, since all the image contains densely populated houses.

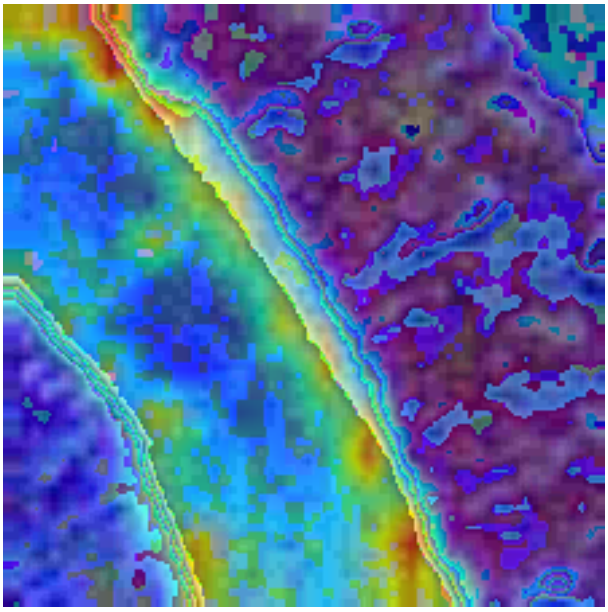

Figure 55: CLS token attention map for Class 8

Class 8 Analysis:

The attention map highlights the strongly distinct edges of the river, thus separating the river water from the banking area.

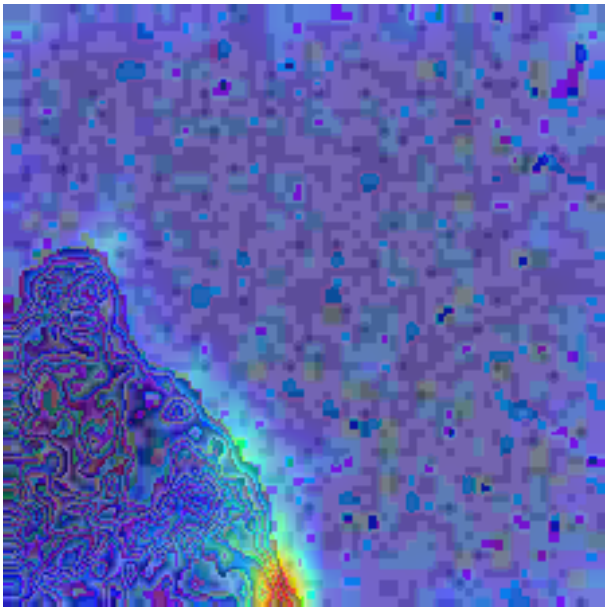

Figure 56: CLS token attention map for Class 9

Class 9 Analysis:

The model, though not strongly differentiating areas, still gives differing shades of importance to the bank and the sea.